

SUPPLEMENTARY MATERIAL

CLASSIFICATION OF INTEGRAL MODULAR DATA UP TO RANK 13

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I. CATALOGUE OF INTEGRAL MODULAR DATA UP TO RANK 13

This section presents a comprehensive catalogue of integral modular data (MD) up to rank 13. Notably, all the MD listed here are categorifiable, meaning each corresponds to the MD of an integral modular fusion category. These corresponding categorical models are identified in [1, Theorem 1.3]. We will begin by listing the non-pointed MD in §I.1, followed by the pointed MD in §I.2. Additionally, these data can be accessed in a machine-readable format in [2].

I.1. Non-Pointed Modular Data Up To Rank 13. The data are organized in the following sequence:

- Rank,
- Type,
- Fusion coefficients,
- List of topological spins (s_i) giving the T-matrix (see [1, Definition 2.11]),
- Cyclotomic conductor of the S -matrix,
- Cyclotomic conductor of the T -matrix (its order),
- Central charge,
- List of 2nd Frobenius-Schur indicators,
- S -matrix.

I.1.1. Rank 8, type $[1, 1, 2, 2, 2, 2, 3, 3]$, first fusion ring.

1 0 0 0 0 0 0 0	0 1 0 0 0 0 0 0	0 0 1 0 0 0 0 0	0 0 0 1 0 0 0 0	0 0 0 0 1 0 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 0 1 0	0 0 0 0 0 0 0 1
0 1 0 0 0 0 0 0	1 0 0 0 0 0 0 0	0 0 1 0 0 0 0 0	0 0 0 1 0 0 0 0	0 0 0 0 1 0 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 0 1 0	0 0 0 0 0 0 0 1
0 0 1 0 0 0 0 0	0 0 0 1 0 0 0 0	1 1 0 0 0 1 0 0	0 0 0 1 1 0 0 0	0 0 0 1 0 1 0 0	0 0 1 0 1 0 0 0	0 0 0 0 0 0 1 1	0 0 0 0 0 0 0 1
0 0 0 1 0 0 0 0	0 0 0 0 1 0 0 0	0 0 0 1 1 0 0 0	1 1 1 0 0 0 0 0	0 0 1 0 0 1 0 0	0 0 0 0 1 1 0 0	0 0 0 0 0 0 1 1	0 0 0 0 0 0 0 1
0 0 0 0 1 0 0 0	0 0 0 0 0 1 0 0	0 0 0 1 0 1 0 0	0 0 1 0 0 1 0 0	1 1 0 0 1 0 0 0	0 0 1 1 0 0 0 0	0 0 0 0 0 0 1 1	0 0 0 0 0 0 0 1
0 0 0 0 0 1 0 0	0 0 0 0 0 0 1 0	0 0 1 0 1 0 0 0	0 0 0 0 1 1 0 0	0 0 1 1 0 0 0 0	1 1 0 1 0 0 0 0	0 0 0 0 0 0 1 1	0 0 0 0 0 0 0 1
0 0 0 0 0 0 1 0	0 0 0 0 0 0 0 1	0 0 0 0 0 0 1 1	0 0 0 0 0 0 1 1	0 0 0 0 0 0 1 1	0 0 0 0 0 0 1 1	1 0 1 1 1 1 0 0	0 1 1 1 1 1 0 0
0 0 0 0 0 0 0 1	0 0 0 0 0 0 0 1	0 0 0 0 0 0 1 1	0 0 0 0 0 0 1 1	0 0 0 0 0 0 1 1	0 0 0 0 0 0 1 1	0 1 1 1 1 1 0 0	1 0 1 1 1 1 0 0

- $[0, 0, -2/9, 4/9, 0, 1/9, 1/2, 0], 9, 18, 0, [1, 1, 1, 1, 1, 1, 1, 1],$

$$\begin{bmatrix}
 1 & 1 & 2 & 2 & 2 & 2 & 3 & 3 \\
 1 & 1 & 2 & 2 & 2 & 2 & -3 & -3 \\
 2 & 2 & 2\zeta_9^4 + 2\zeta_9^5 & 2\zeta_9^2 + 2\zeta_9^7 & -2 & 2\zeta_9 + 2\zeta_9^8 & 0 & 0 \\
 2 & 2 & 2\zeta_9^2 + 2\zeta_9^7 & 2\zeta_9 + 2\zeta_9^8 & -2 & 2\zeta_9^4 + 2\zeta_9^5 & 0 & 0 \\
 2 & 2 & -2 & -2 & 4 & -2 & 0 & 0 \\
 2 & 2 & 2\zeta_9 + 2\zeta_9^8 & 2\zeta_9^4 + 2\zeta_9^5 & -2 & 2\zeta_9^2 + 2\zeta_9^7 & 0 & 0 \\
 3 & -3 & 0 & 0 & 0 & 0 & 3 & -3 \\
 3 & -3 & 0 & 0 & 0 & 0 & -3 & 3
 \end{bmatrix},$$

- $[0, 0, -1/3, 1/3, 0, 0, -1/4, 1/4], 1, 12, 0, [1, 1, 1, 1, 1, 1, -1, -1].$

$$\begin{bmatrix} 1 & 1 & 2 & 2 & 2 & 2 & 3 & 3 \\ 1 & 1 & 2 & 2 & 2 & 2 & -3 & -3 \\ 2 & 2 & -2 & 4 & -2 & -2 & 0 & 0 \\ 2 & 2 & 4 & -2 & -2 & -2 & 0 & 0 \\ 2 & 2 & -2 & -2 & 4 & -2 & 0 & 0 \\ 2 & 2 & -2 & -2 & -2 & 4 & 0 & 0 \\ 3 & -3 & 0 & 0 & 0 & 0 & -3 & 3 \\ 3 & -3 & 0 & 0 & 0 & 0 & 3 & -3 \end{bmatrix}$$

- $[0, 0, -1/3, -1/3, 1/3, 1/3, -1/4, 1/4], 1, 12, 4, [1, 1, 1, 1, 1, 1, 1, 1],$

$$\begin{bmatrix} 1 & 1 & 2 & 2 & 2 & 2 & 3 & 3 \\ 1 & 1 & 2 & 2 & 2 & 2 & -3 & -3 \\ 2 & 2 & -2 & 4 & -2 & -2 & 0 & 0 \\ 2 & 2 & 4 & -2 & -2 & -2 & 0 & 0 \\ 2 & 2 & -2 & -2 & -2 & 4 & 0 & 0 \\ 2 & 2 & -2 & -2 & 4 & -2 & 0 & 0 \\ 3 & -3 & 0 & 0 & 0 & 0 & 3 & -3 \\ 3 & -3 & 0 & 0 & 0 & 0 & -3 & 3 \end{bmatrix}$$

I.1.3. *Rank 10, type $[1, 1, 1, 2, 2, 2, 2, 2, 2, 3]$, first fusion ring.*

[illegible]

- $[0, 0, 0, 1/9, 4/9, -2/9, -2/9, 4/9, 1/9, 1/2], 9, 18, 4, [1, 0, 0, 0, 0, 0, 0, 0, 0, 1].$

1	1	1	2	2	2	2	2	2	3
1	1	1	$2\zeta_3^2$	$2\zeta_3$	$2\zeta_3$	$2\zeta_3^2$	$2\zeta_3^2$	$2\zeta_3$	3
1	1	1	$2\zeta_3$	$2\zeta_3^2$	$2\zeta_3^2$	$2\zeta_3$	$2\zeta_3$	$2\zeta_3^2$	3
2	$2\zeta_3^2$	$2\zeta_3$	$-2\zeta_9^7$	$-2\zeta_9^5$	$2\zeta_9^2 + 2\zeta_9^5$	$2\zeta_9^4 + 2\zeta_9^7$	$-2\zeta_9^4$	$-2\zeta_9^2$	0
2	$2\zeta_3$	$2\zeta_3^2$	$-2\zeta_9^5$	$2\zeta_9^4 + 2\zeta_9^7$	$-2\zeta_9^7$	$-2\zeta_9^2$	$2\zeta_9^2 + 2\zeta_9^5$	$-2\zeta_9^4$	0
2	$2\zeta_3$	$2\zeta_3^2$	$2\zeta_9^2 + 2\zeta_9^5$	$-2\zeta_9^7$	$-2\zeta_9^4$	$-2\zeta_9^5$	$-2\zeta_9^2$	$2\zeta_9^4 + 2\zeta_9^7$	0
2	$2\zeta_3^2$	$2\zeta_3$	$2\zeta_9^4 + 2\zeta_9^7$	$-2\zeta_9^2$	$-2\zeta_9^5$	$-2\zeta_9^4$	$-2\zeta_9^7$	$2\zeta_9^2 + 2\zeta_9^5$	0
2	$2\zeta_3^2$	$2\zeta_3$	$-2\zeta_9^4$	$2\zeta_9^2 + 2\zeta_9^5$	$-2\zeta_9^2$	$-2\zeta_9^7$	$2\zeta_9^4 + 2\zeta_9^7$	$-2\zeta_9^5$	0
2	$2\zeta_3$	$2\zeta_3^2$	$-2\zeta_9^2$	$-2\zeta_9^4$	$2\zeta_9^4 + 2\zeta_9^7$	$2\zeta_9^2 + 2\zeta_9^5$	$-2\zeta_9^5$	$-2\zeta_9^7$	0
3	3	3	0	0	0	0	0	0	-3

- $[0, 0, 0, 2/9, -1/9, -4/9, -4/9, -1/9, 2/9, 1/2], 9, 18, 4, [1, 0, 0, 0, 0, 0, 0, 0, 0, 1],$

1	1	1	2	2	2	2	2	2	3
1	1	1	$2\zeta_3$	$2\zeta_3^2$	$2\zeta_3^2$	$2\zeta_3$	$2\zeta_3$	$2\zeta_3^2$	3
1	1	1	$2\zeta_3^2$	$2\zeta_3$	$2\zeta_3$	$2\zeta_3^2$	$2\zeta_3^2$	$2\zeta_3$	3
2	$2\zeta_3$	$2\zeta_3^2$	$-2\zeta_9^5$	$2\zeta_9^4 + 2\zeta_9^7$	$-2\zeta_9^7$	$-2\zeta_9^2$	$2\zeta_9^2 + 2\zeta_9^5$	$-2\zeta_9^4$	0
2	$2\zeta_3^2$	$2\zeta_3$	$2\zeta_9^4 + 2\zeta_9^7$	$-2\zeta_9^2$	$-2\zeta_9^5$	$-2\zeta_9^4$	$-2\zeta_9^7$	$2\zeta_9^2 + 2\zeta_9^5$	0
2	$2\zeta_3^2$	$2\zeta_3$	$-2\zeta_9^7$	$-2\zeta_9^5$	$2\zeta_9^2 + 2\zeta_9^5$	$2\zeta_9^4 + 2\zeta_9^7$	$-2\zeta_9^4$	$-2\zeta_9^2$	0
2	$2\zeta_3$	$2\zeta_3^2$	$-2\zeta_9^2$	$-2\zeta_9^4$	$2\zeta_9^4 + 2\zeta_9^7$	$2\zeta_9^2 + 2\zeta_9^5$	$-2\zeta_9^5$	$-2\zeta_9^7$	0
2	$2\zeta_3$	$2\zeta_3^2$	$2\zeta_9^2 + 2\zeta_9^5$	$-2\zeta_9^7$	$-2\zeta_9^4$	$-2\zeta_9^5$	$-2\zeta_9^2$	$2\zeta_9^4 + 2\zeta_9^7$	0
2	$2\zeta_3^2$	$2\zeta_3$	$-2\zeta_9^4$	$2\zeta_9^2 + 2\zeta_9^5$	$-2\zeta_9^2$	$-2\zeta_9^7$	$2\zeta_9^4 + 2\zeta_9^7$	$-2\zeta_9^5$	0
3	3	3	0	0	0	0	0	0	-3

[illegible][illegible]

- $$\begin{bmatrix} 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 & 3 \\ 1 & 1 & 1 & 2\zeta_3^2 & 2\zeta_3 & 2\zeta_3 & 2\zeta_3^2 & 2\zeta_3^2 & 2\zeta_3 & 3 \\ 1 & 1 & 1 & 2\zeta_3 & 2\zeta_3^2 & 2\zeta_3^2 & 2\zeta_3 & 2\zeta_3 & 2\zeta_3^2 & 3 \\ 2 & 2\zeta_3^2 & 2\zeta_3 & -2 & -2\zeta_3 & -2\zeta_3^2 & -2\zeta_3 & -2\zeta_3^2 & -2 & 0 \\ 2 & 2\zeta_3 & 2\zeta_3^2 & -2\zeta_3 & -2\zeta_3 & -2 & -2 & -2\zeta_3^2 & -2\zeta_3^2 & 0 \\ 2 & 2\zeta_3 & 2\zeta_3^2 & -2\zeta_3^2 & -2 & -2\zeta_3^2 & -2\zeta_3 & -2 & -2\zeta_3 & 0 \\ 2 & 2\zeta_3^2 & 2\zeta_3 & -2\zeta_3 & -2 & -2\zeta_3 & -2\zeta_3^2 & -2 & -2\zeta_3^2 & 0 \\ 2 & 2\zeta_3^2 & 2\zeta_3 & -2\zeta_3^2 & -2\zeta_3^2 & -2 & -2 & -2\zeta_3 & -2\zeta_3 & 0 \\ 2 & 2\zeta_3 & 2\zeta_3^2 & -2 & -2\zeta_3^2 & -2\zeta_3 & -2\zeta_3^2 & -2\zeta_3 & -2 & 0 \\ 3 & 3 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & -3 \end{bmatrix}$$

[illegible][illegible]

- | | | | | | | | | | | |
|---|----|----|----|--------------------------|--------------------------|--------------------------|----|--------------------------|--------------------------|--------------------------|
| 1 | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 1 | 1 | 1 | 1 | -2 | -2 | 2 | 2 | 2 | -2 | -2 |
| 1 | 1 | 1 | 1 | -2 | 2 | -2 | 2 | -2 | 2 | -2 |
| 1 | 1 | 1 | 1 | 2 | -2 | -2 | 2 | -2 | -2 | 2 |
| 2 | -2 | -2 | 2 | $2\zeta_8 - 2\zeta_8^3$ | 0 | 0 | 0 | 0 | 0 | $-2\zeta_8 + 2\zeta_8^3$ |
| 2 | -2 | 2 | -2 | 0 | $-2\zeta_8 + 2\zeta_8^3$ | 0 | 0 | 0 | $2\zeta_8 - 2\zeta_8^3$ | 0 |
| 2 | 2 | -2 | -2 | 0 | 0 | $-2\zeta_8 + 2\zeta_8^3$ | 0 | $2\zeta_8 - 2\zeta_8^3$ | 0 | 0 |
| 2 | 2 | 2 | 2 | 0 | 0 | 0 | -4 | 0 | 0 | 0 |
| 2 | 2 | -2 | -2 | 0 | 0 | $2\zeta_8 - 2\zeta_8^3$ | 0 | $-2\zeta_8 + 2\zeta_8^3$ | 0 | 0 |
| 2 | -2 | 2 | -2 | 0 | $2\zeta_8 - 2\zeta_8^3$ | 0 | 0 | 0 | $-2\zeta_8 + 2\zeta_8^3$ | 0 |
| 2 | -2 | -2 | 2 | $-2\zeta_8 + 2\zeta_8^3$ | 0 | 0 | 0 | 0 | 0 | $2\zeta_8 - 2\zeta_8^3$ |

- $[0, 0, 0, 0, -7/16, -3/16, 1/16, 1/4, -7/16, 5/16, 1/16]$, 8, 16, 1, $[1, 1, 1, 1, -1, 1, -1, 1, -1, 1, -1]$,

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & -2 & -2 & 2 & 2 & 2 & -2 & -2 \\ 1 & 1 & 1 & 1 & -2 & 2 & -2 & 2 & -2 & 2 & -2 \\ 1 & 1 & 1 & 1 & 2 & -2 & -2 & 2 & -2 & -2 & 2 \\ 2 & -2 & -2 & 2 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 & 0 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 \\ 2 & -2 & 2 & -2 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 \\ 2 & 2 & -2 & -2 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 \\ 2 & 2 & 2 & 2 & 0 & 0 & 0 & -4 & 0 & 0 & 0 \\ 2 & 2 & -2 & -2 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 \\ 2 & -2 & 2 & -2 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 \\ 2 & -2 & -2 & 2 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 & 0 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 \end{bmatrix}$$

- $[0, 0, 0, 0, -7/16, 1/16, 1/16, 1/4, -7/16, -7/16, 1/16]$, 8, 16, 1, $[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]$,

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & -2 & -2 & 2 & 2 & 2 & -2 & -2 \\ 1 & 1 & 1 & 1 & -2 & 2 & -2 & 2 & -2 & 2 & -2 \\ 1 & 1 & 1 & 1 & 2 & -2 & -2 & 2 & -2 & -2 & 2 \\ 2 & -2 & -2 & 2 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 & 0 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 \\ 2 & -2 & 2 & -2 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 \\ 2 & 2 & -2 & -2 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 \\ 2 & 2 & 2 & 2 & 0 & 0 & 0 & -4 & 0 & 0 & 0 \\ 2 & 2 & -2 & -2 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 \\ 2 & -2 & 2 & -2 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 \\ 2 & -2 & -2 & 2 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 & 0 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 \end{bmatrix}$$

- $[0, 0, 0, 0, -3/16, 5/16, 5/16, 1/4, -3/16, -3/16, 5/16], 8, 16, 1, [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & -2 & -2 & 2 & 2 & 2 & -2 & -2 \\ 1 & 1 & 1 & 1 & -2 & 2 & -2 & 2 & -2 & 2 & -2 \\ 1 & 1 & 1 & 1 & 2 & -2 & -2 & 2 & -2 & -2 & 2 \\ 2 & -2 & -2 & 2 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 & 0 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 \\ 2 & -2 & 2 & -2 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 \\ 2 & 2 & -2 & -2 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 \\ 2 & 2 & 2 & 2 & 0 & 0 & 0 & -4 & 0 & 0 & 0 \\ 2 & 2 & -2 & -2 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 \\ 2 & -2 & 2 & -2 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 \\ 2 & -2 & -2 & 2 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 & 0 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 \end{bmatrix}$$

- $[0, 0, 0, 0, -5/16, 7/16, 7/16, -1/4, -1/16, -1/16, 3/16], 8, 16, -1, [1, 1, 1, 1, 1, -1, -1, 1, -1, -1, 1],$

1	1	1	1	2	2	2	2	2	2	2
1	1	1	1	-2	-2	2	2	2	-2	-2
1	1	1	1	-2	2	-2	2	-2	2	-2
1	1	1	1	2	-2	-2	2	-2	-2	2
2	-2	-2	2	$-2\zeta_8 + 2\zeta_8^3$	0	0	0	0	0	$2\zeta_8 - 2\zeta_8^3$
2	-2	2	-2	0	$2\zeta_8 - 2\zeta_8^3$	0	0	0	$-2\zeta_8 + 2\zeta_8^3$	0
2	2	-2	-2	0	0	$2\zeta_8 - 2\zeta_8^3$	0	$-2\zeta_8 + 2\zeta_8^3$	0	0
2	2	2	2	0	0	0	-4	0	0	0
2	2	-2	-2	0	0	$-2\zeta_8 + 2\zeta_8^3$	0	$2\zeta_8 - 2\zeta_8^3$	0	0
2	-2	2	-2	0	$-2\zeta_8 + 2\zeta_8^3$	0	0	0	$2\zeta_8 - 2\zeta_8^3$	0
2	-2	-2	2	$2\zeta_8 - 2\zeta_8^3$	0	0	0	0	0	$-2\zeta_8 + 2\zeta_8^3$

- $[0, 0, 0, 0, -5/16, 7/16, -5/16, -1/4, 3/16, -1/16, 3/16], 8, 16, -1, [1, 1, 1, 1, -1, 1, -1, 1, -1, 1, -1],$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & -2 & -2 & 2 & 2 & 2 & -2 & -2 & -2 \\ 1 & 1 & 1 & 1 & -2 & 2 & -2 & 2 & -2 & 2 & -2 & -2 \\ 1 & 1 & 1 & 1 & 2 & -2 & -2 & 2 & -2 & -2 & 2 & 2 \\ 2 & -2 & -2 & 2 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 & 0 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 \\ 2 & -2 & 2 & -2 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 \\ 2 & 2 & -2 & -2 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 & 0 \\ 2 & 2 & 2 & 2 & 0 & 0 & 0 & -4 & 0 & 0 & 0 & 0 \\ 2 & 2 & -2 & -2 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 & 0 \\ 2 & -2 & 2 & -2 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 \\ 2 & -2 & -2 & 2 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 & 0 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 \end{bmatrix}$$

- $[0, 0, 0, 0, -5/16, -5/16, -5/16, -1/4, 3/16, 3/16, 3/16], 8, 16, -1, [1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1],$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & -2 & -2 & 2 & 2 & 2 & -2 & -2 & -2 \\ 1 & 1 & 1 & 1 & -2 & 2 & -2 & 2 & -2 & 2 & -2 & -2 \\ 1 & 1 & 1 & 1 & 2 & -2 & -2 & 2 & -2 & -2 & 2 & 2 \\ 2 & -2 & -2 & 2 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 & 0 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 \\ 2 & -2 & 2 & -2 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 \\ 2 & 2 & -2 & -2 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 & 0 \\ 2 & 2 & 2 & 2 & 0 & 0 & 0 & -4 & 0 & 0 & 0 & 0 \\ 2 & 2 & -2 & -2 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 & 0 \\ 2 & -2 & 2 & -2 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 \\ 2 & -2 & -2 & 2 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 & 0 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 \end{bmatrix}$$

- $[0, 0, 0, 0, -1/16, -1/16, -1/16, -1/4, 7/16, 7/16, 7/16], 8, 16, -1, [1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1],$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & -2 & -2 & 2 & 2 & 2 & -2 & -2 & -2 \\ 1 & 1 & 1 & 1 & -2 & 2 & -2 & 2 & -2 & 2 & -2 & -2 \\ 1 & 1 & 1 & 1 & 2 & -2 & -2 & 2 & -2 & -2 & 2 & 2 \\ 2 & -2 & -2 & 2 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 & 0 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 \\ 2 & -2 & 2 & -2 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 \\ 2 & 2 & -2 & -2 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 & 0 \\ 2 & 2 & 2 & 2 & 0 & 0 & 0 & -4 & 0 & 0 & 0 & 0 \\ 2 & 2 & -2 & -2 & 0 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 & 0 \\ 2 & -2 & 2 & -2 & 0 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 & 0 \\ 2 & -2 & -2 & 2 & -2\zeta_8 + 2\zeta_8^3 & 0 & 0 & 0 & 0 & 0 & 2\zeta_8 - 2\zeta_8^3 & 0 \end{bmatrix}$$

I.2. Pointed Modular Data Up To Rank 13. In light of [1, Remark 1.2], it suffices to specify only the Abelian groups and their corresponding T-matrices, which encapsulate the topological spins in this context. Let us denote by C_n the cyclic group of order n . The complete data set, akin to the format presented in §I.1, is available at [2].

- C_1 : $[0]$,
- C_2 : $[0, 1/4], [0, -1/4]$,
- C_3 : $[0, 1/3, 1/3], [0, -1/3, -1/3]$,
- C_2^2 : $[0, 1/2, 0, 0], [0, -1/4, 1/4, 0], [0, 1/2, 1/4, 1/4], [0, 1/2, 1/2, 1/2], [0, -1/4, -1/4, 1/2]$,
- C_4 : $[0, 1/2, 1/8, 1/8], [0, 1/2, 3/8, 3/8], [0, 1/2, -3/8, -3/8], [0, 1/2, -1/8, -1/8]$,
- C_5 : $[0, 1/5, -1/5, -1/5, 1/5], [0, 2/5, -2/5, -2/5, 2/5]$,
- C_6 : $[0, -1/4, 1/12, 1/3, 1/3, 1/12], [0, -1/4, 5/12, -1/3, -1/3, 5/12], [0, 1/4, -5/12, 1/3, 1/3, -5/12], [0, 1/4, -1/12, -1/3, -1/3, -1/12]$,
- C_7 : $[0, 1/7, -3/7, 2/7, 2/7, -3/7, 1/7], [0, 3/7, -2/7, -1/7, -1/7, -2/7, 3/7]$,
- C_2^3 : $[0, -1/4, 1/4, 1/4, 1/4, 1/2, 0, 0], [0, -1/4, -1/4, -1/4, 1/4, 1/2, 0, 0], [0, -1/4, 1/2, 1/2, 1/4, 1/2, 1/4, 1/4], [0, -1/4, 1/2, -1/4, 1/2, -1/4, 1/2, 1/4]$,
- $C_2 \times C_4$: $[0, 1/2, -1/4, 1/4, 1/8, 3/8, 3/8, 1/8], [0, 1/2, 1/4, -1/4, 1/8, -1/8, -1/8, 1/8], [0, 1/2, -1/4, 1/4, 3/8, -3/8, -3/8, 3/8], [0, 1/2, -1/4, 1/4, -3/8, -1/8, -1/8, -3/8]$,
- C_8 : $[0, 0, 1/4, -7/16, 1/16, 1/16, -7/16, 1/4], [0, 0, 1/4, -3/16, 5/16, 5/16, -3/16, 1/4], [0, 0, -1/4, -5/16, 3/16, 3/16, -5/16, -1/4], [0, 0, -1/4, -1/16, 7/16, 7/16, -1/16, -1/4]$,
- C_3^2 : $[0, 0, 0, 1/3, -1/3, -1/3, 1/3, 0, 0], [0, 1/3, 1/3, -1/3, -1/3, -1/3, 1/3, 1/3]$,
- C_9 : $[0, 0, 1/9, -2/9, 4/9, 4/9, -2/9, 0], [0, 0, 2/9, -4/9, -1/9, -1/9, -4/9, 2/9, 0]$,
- C_{10} : $[0, 1/4, 1/20, -1/5, 1/5, 9/20, 9/20, 1/5, -1/5, 1/20], [0, -1/4, 3/20, 2/5, -2/5, 7/20, 7/20, -2/5, 2/5, 3/20], [0, -1/4, -9/20, -1/5, 1/5, -1/20, -1/20, 1/5, -1/5, -9/20], [0, 1/4, -7/20, 2/5, -2/5, -3/20, -3/20, -2/5, 2/5, -7/20]$,

1	1	1	3	3	3	3	3	3	3	3	3	5	5	5	5	5	5
1	1	1	3	3	3	3	3	3	3	3	3	5 ζ_3	5 ζ_3^2	5 ζ_3	5 ζ_3^2	5 ζ_3	5 ζ_3^2
1	1	1	3	3	3	3	3	3	3	3	3	5 ζ_3^2	5 ζ_3	5 ζ_3^2	5 ζ_3	5 ζ_3^2	5 ζ_3
3	3	3	6 $\zeta_3^3 + 3\zeta_3^4$	3 $\zeta_3 + 6\zeta_3^2$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	3 $\zeta_3^2 + 6\zeta_3^4$	6 $\zeta_3 + 3\zeta_3^3$	0	0	0	0	0	0	0
3	3	3	3 $\zeta_3 + 6\zeta_3^2$	6 $\zeta_3^2 + 3\zeta_3^4$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	6 $\zeta_3 + 3\zeta_3^3$	3 $\zeta_3^2 + 6\zeta_3^4$	0	0	0	0	0	0	0
3	3	3	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	3 $\zeta_3^2 + 6\zeta_3^4$	6 $\zeta_3 + 3\zeta_3^3$	6 $\zeta_3^3 + 3\zeta_3^4$	3 $\zeta_3 + 6\zeta_3^2$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	0	0	0	0	0	0	0
3	3	3	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	6 $\zeta_3 + 3\zeta_3^3$	3 $\zeta_3^2 + 6\zeta_3^4$	6 $\zeta_3^3 + 3\zeta_3^4$	3 $\zeta_3 + 6\zeta_3^2$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	0	0	0	0	0	0	0
3	3	3	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	6 $\zeta_3^3 + 3\zeta_3^4$	3 $\zeta_3 + 6\zeta_3^2$	6 $\zeta_3 + 3\zeta_3^3$	3 $\zeta_3 + 6\zeta_3^2$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	0	0	0	0	0	0	0
3	3	3	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	3 $\zeta_3 + 6\zeta_3^2$	6 $\zeta_3^3 + 3\zeta_3^4$	3 $\zeta_3^2 + 6\zeta_3^4$	6 $\zeta_3 + 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	0	0	0	0	0	0	0
3	3	3	3 $\zeta_3^2 + 6\zeta_3^4$	6 $\zeta_3 + 3\zeta_3^3$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	3 $\zeta_3 + 6\zeta_3^2$	6 $\zeta_3^3 + 3\zeta_3^4$	0	0	0	0	0	0	0
3	3	3	6 $\zeta_3 + 3\zeta_3^3$	3 $\zeta_3^2 + 6\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	6 $\zeta_3^3 + 3\zeta_3^4$	3 $\zeta_3 + 6\zeta_3^2$	0	0	0	0	0	0	0
5	5 ζ_3	5 ζ_3^2	0	0	0	0	0	0	0	0	-5 ζ_3^4	-5 ζ_3^5	5 $\zeta_3^4 + 5\zeta_3^7$	5 $\zeta_3^2 + 5\zeta_3^6$	-5 ζ_3^7	-5 ζ_3^6	-5 ζ_3^5
5	5 ζ_3^2	5 ζ_3	0	0	0	0	0	0	0	0	-5 ζ_3^5	-5 ζ_3^6	5 $\zeta_3^4 + 5\zeta_3^7$	5 $\zeta_3^2 + 5\zeta_3^6$	-5 ζ_3^7	-5 ζ_3^6	-5 ζ_3^5
5	5 ζ_3	5 ζ_3^2	0	0	0	0	0	0	0	0	5 $\zeta_3^4 + 5\zeta_3^7$	5 $\zeta_3^2 + 5\zeta_3^6$	-5 ζ_3^7	-5 ζ_3^6	-5 ζ_3^5	-5 ζ_3^4	-5 ζ_3^3
5	5 ζ_3^2	5 ζ_3	0	0	0	0	0	0	0	0	5 $\zeta_3^2 + 5\zeta_3^6$	5 $\zeta_3^4 + 5\zeta_3^7$	-5 ζ_3^7	-5 ζ_3^6	-5 ζ_3^5	-5 ζ_3^4	-5 ζ_3^3
5	5 ζ_3	5 ζ_3^2	0	0	0	0	0	0	0	0	-5 ζ_3^7	-5 ζ_3^6	-5 ζ_3^5	-5 ζ_3^4	-5 ζ_3^3	-5 ζ_3^2	-5 ζ_3
5	5 ζ_3^2	5 ζ_3	0	0	0	0	0	0	0	0	-5 ζ_3^6	-5 ζ_3^5	-5 ζ_3^4	-5 ζ_3^3	-5 ζ_3^2	-5 ζ_3	-5 ζ_3^0

• $[0, 0, 0, -2/5, -2/5, -1/5, -1/5, 1/5, 1/5, 2/5, 2/5, -4/9, -4/9, -1/9, -1/9, 2/9, 2/9], 45, 45, 4, [1, 0, \dots, 0],$

1	1	1	3	3	3	3	3	3	3	3	3	5	5	5	5	5	5
1	1	1	3	3	3	3	3	3	3	3	3	5 ζ_3^2	5 ζ_3	5 ζ_3^2	5 ζ_3	5 ζ_3^2	5 ζ_3
1	1	1	3	3	3	3	3	3	3	3	3	5 ζ_3	5 ζ_3^2	5 ζ_3	5 ζ_3^2	5 ζ_3	5 ζ_3^2
3	3	3	6 $\zeta_3^3 + 3\zeta_3^4$	3 $\zeta_3 + 6\zeta_3^2$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	3 $\zeta_3^2 + 6\zeta_3^4$	6 $\zeta_3 + 3\zeta_3^3$	0	0	0	0	0	0	0
3	3	3	3 $\zeta_3 + 6\zeta_3^2$	6 $\zeta_3^2 + 3\zeta_3^4$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	6 $\zeta_3 + 3\zeta_3^3$	3 $\zeta_3^2 + 6\zeta_3^4$	0	0	0	0	0	0	0
3	3	3	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	3 $\zeta_3^2 + 6\zeta_3^4$	6 $\zeta_3 + 3\zeta_3^3$	6 $\zeta_3^3 + 3\zeta_3^4$	3 $\zeta_3 + 6\zeta_3^2$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	0	0	0	0	0	0	0
3	3	3	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	6 $\zeta_3 + 3\zeta_3^3$	3 $\zeta_3^2 + 6\zeta_3^4$	6 $\zeta_3^3 + 3\zeta_3^4$	3 $\zeta_3 + 6\zeta_3^2$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	0	0	0	0	0	0	0
3	3	3	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	6 $\zeta_3^3 + 3\zeta_3^4$	3 $\zeta_3 + 6\zeta_3^2$	6 $\zeta_3 + 3\zeta_3^3$	3 $\zeta_3 + 6\zeta_3^2$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	0	0	0	0	0	0	0
3	3	3	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	3 $\zeta_3 + 6\zeta_3^2$	6 $\zeta_3^3 + 3\zeta_3^4$	3 $\zeta_3^2 + 6\zeta_3^4$	6 $\zeta_3 + 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	0	0	0	0	0	0	0
3	3	3	3 $\zeta_3^2 + 6\zeta_3^4$	6 $\zeta_3 + 3\zeta_3^3$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	3 $\zeta_3 + 6\zeta_3^2$	6 $\zeta_3^3 + 3\zeta_3^4$	0	0	0	0	0	0	0
3	3	3	6 $\zeta_3 + 3\zeta_3^3$	3 $\zeta_3^2 + 6\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3 - 3\zeta_3^4$	-3 $\zeta_3^2 - 3\zeta_3^3$	-3 $\zeta_3^2 - 3\zeta_3^3$	6 $\zeta_3^3 + 3\zeta_3^4$	3 $\zeta_3 + 6\zeta_3^2$	0	0	0	0	0	0	0
5	5 ζ_3^2	5 ζ_3	0	0	0	0	0	0	0	0	5 $\zeta_3^2 + 5\zeta_3^6$	5 $\zeta_3^4 + 5\zeta_3^7$	-5 ζ_3^5	-5 ζ_3^4	-5 ζ_3^3	-5 ζ_3^2	-5 ζ_3
5	5 ζ_3	5 ζ_3^2	0	0	0	0	0	0	0	0	5 $\zeta_3^4 + 5\zeta_3^7$	5 $\zeta_3^2 + 5\zeta_3^6$	-5 ζ_3^7	-5 ζ_3^6	-5 ζ_3^5	-5 ζ_3^4	-5 ζ_3^3
5	5 ζ_3^2	5 ζ_3	0	0	0	0	0	0	0	0	-5 ζ_3^7	-5 ζ_3^6	-5 ζ_3^5	-5 ζ_3^4	-5 ζ_3^3	-5 ζ_3^2	-5 ζ_3
5	5 ζ_3	5 ζ_3^2	0	0	0	0	0	0	0	0	-5 ζ_3^6	-5 ζ_3^5	-5 ζ_3^4	-5 ζ_3^3	-5 ζ_3^2	-5 ζ_3	-5 ζ_3^0
5	5 ζ_3^2	5 ζ_3	0	0	0	0	0	0	0	0	-5 ζ_3^5	-5 ζ_3^4	-5 ζ_3^3	-5 ζ_3^2	-5 ζ_3	-5 ζ_3^0	-5 ζ_3^{-1}
5	5 ζ_3	5 ζ_3^2	0	0	0	0	0	0	0	0	-5 ζ_3^4	-5 ζ_3^3	-5 ζ_3^2	-5 ζ_3	-5 ζ_3^0	-5 ζ_3^{-1}	-5 ζ_3^{-2}

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